

Learning-Aware Lyapunov Resource Control for Correlated Multi-Agent Markov Learning

Anonymous Authors

Abstract—Parallel Markov trajectories can accelerate policy evaluation, but their value depends on cross-agent correlation, temporal persistence, communication, and update staleness. We formulate allocation as finite-budget learning-value control: a server selects participation, sample spacing, and step size before an update. A discrete Lyapunov certificate converts these decisions into a terminal-risk objective for affine delayed temporal-difference learning. Its terms retain cross-agent second moments, finite-mixing bias, and the delay profile, so the certificate both proves convergence and designs the resource action. When dependence is unknown, a learning-aware gate purchases a fully charged likelihood probe only when its downstream risk reduction exceeds message, transition, delay, and decision-error costs. We establish a stopped adaptive change-of-measure lower bound, a finite-horizon performance and safety theorem, and a necessary-sufficient budget-threshold sandwich on a separated Markov class. Controlled CPU studies recover correlation-limited speedup, calibrate the finite-time bounds, and reduce two finite-budget risks by 54.4 and 10.9 percent. On 64 held-out seeds, joint control reduces parameter-error and discounted-return-estimation AUC by 9.9 and 11.3 percent against a development-selected strong fixed baseline. Held-out multi-agent proximal policy optimization (MAPPO) studies on particle coordination and continuous-control locomotion select dependence-matched participation in all 32 controller runs. Under identical charged budgets, the controller improves equal-mixture team-return AUC by 2.32 and 8.34 percent over each task’s strongest single fixed-participation comparator.

Index Terms—Multi-agent learning, Markov data, temporal-difference learning, adaptive sensing, Lyapunov methods.

I. INTRODUCTION

Policy evaluation asks how much discounted return a fixed policy will obtain in a Markov decision process. The state describes the physical or informational configuration that determines the next reward and transition, the target policy maps each state to an action distribution, and the value function records the expected discounted return under repeated execution of that policy. Accurate evaluation supplies critics, advantages, and policy-improvement directions in actor-critic and centralized-training/decentralized-execution systems, and it is also fundamental to parallel simulation and federated or on-device reinforcement learning.

We study the common architecture in which several actors execute the same target policy on parallel Markov environment copies and a server aggregates their temporal-difference observations into one value parameter. At each learning block, the server can select the number q of participating actors, advance each selected chain by b physical transitions before retaining an observation, and apply step size η to the aggregate update. These choices consume two noninterchangeable resources: actor messages and environment transitions. The central question

is therefore not simply how to parallelize an update, but how to convert a finite spatial-temporal sampling budget into the largest certified reduction in learning risk.

This allocation is difficult because parallel samples need not provide parallel information. Actors may share simulator randomness, exogenous disturbances, replay content, or common environment factors while each actor individually follows the correct target-policy transition law. For example, averaging q equicorrelated Gaussian observations with correlation ρ has effective participation $q/[1 + (q-1)\rho]$, which saturates at $1/\rho$ instead of growing linearly. Broad participation can then spend more messages and shorten the remaining learning horizon without delivering commensurate variance reduction.

Temporal dependence creates the opposing resource effect. Larger spacing b suppresses conditional Markov bias but consumes environment transitions, whereas larger η accelerates transient contraction but raises the stochastic residual floor and can destabilize delayed updates. Participation, spacing, and step size must consequently be chosen together: the best instantaneous variance is generally not the best terminal value estimate under finite budgets. When cross-agent dependence is initially unknown, identifying it is itself a charged action, so information is useful only if the remaining learner can amortize its acquisition cost.

Classical temporal-difference learning and stochastic approximation establish the fixed-point foundations of policy evaluation [1]–[4]. Their modern finite-time analyses quantify how constant steps, Markov mixing, and value-function geometry determine error [5]–[8], while Lyapunov arguments have motivated adaptive learning rates in two-time-scale reinforcement learning [9]. These results analyze a prescribed observation stream; they do not use the finite-time bound to choose the spatial participation and physical sampling clock that create that stream.

Distributed and federated reinforcement learning instead emphasize consensus, parallel sampling, heterogeneous coverage, and sample-communication tradeoffs [10]–[15]. Client-selection, adaptive-batch, antithetic-sampling, and feedback-control mechanisms further regulate which workers participate or how much data they contribute [16]–[20]. These methods optimize participation, update variance, or communication time, but do not jointly retain cross-agent covariance, temporal decorrelation, stable step adaptation, and the action-dependent number of affordable future updates in one learning-risk objective.

Asynchronous stochastic optimization addresses the complementary problem of stale computation and communication [21]–[23]. Delay-adaptive stochastic approximation gives

convergence guarantees under Markovian sampling [24], [25], and an SDDE framework can optimize worker activation and staleness policies for asynchronous stochastic gradient descent [26]. Those formulations principally optimize delay or runtime behavior after specifying the data law. They do not determine whether an additional correlated Markov trajectory is worth its message, transition, and downstream noise cost.

Sequential experimental design, controlled sensing, and best-arm identification provide principled change-of-measure and allocation tools for learning an unknown statistical instance [27]–[33]. Conservative bandits additionally constrain performance relative to a baseline [34]. Their objective is identification or reward, rather than the post-identification risk of a finite-horizon Markov learner. Taken together, the existing classes leave one precise gap: no single mechanism certifies when spatial participation, temporal decorrelation, stable adaptation, and dependence identification improve the final learned value under the same charged budgets.

We close this gap with learning-aware Lyapunov resource control. Before each learning block, the server selects

$$a = (q, b, \eta), \quad (1)$$

where q and b are physical integer decisions and η is a continuous adaptation decision. An exact discrete-time Lyapunov drift bound maps each action to its contraction coefficient and its correlation-, mixing-, and delay-dependent residual floor. Composing the drift over the number of updates affordable under that action produces a certified terminal-risk score, so the Lyapunov function is both the proof certificate and the design objective. For every discrete (q, b) pair, the controller solves a bounded scalar problem in η and then takes the exact finite-catalogue minimizer; it uses neither matrix inversion nor preconditioning.

Unknown dependence introduces a second decision: whether learning the appropriate resource action is worth paying for. The proposed gate purchases a likelihood probe only when the certified downstream improvement of the instance-specific action exceeds the complete message, transition, delay, and decision-error cost. This separates an information boundary, below which reliable identification is impossible, from a learning-value boundary, below which identification is possible but cannot yet improve the learner. Above the learning-value boundary, probing and commitment achieve a finite-budget performance guarantee while respecting an explicit low-instance risk allowance. The same terminal-risk objective thereby connects stochastic optimization, Lyapunov control, and controlled sensing.

The main contributions of this paper are summarized as follows.

- **Finite-Budget Learning-Value Formulation:** We formulate correlated multi-agent policy evaluation as joint control of spatial participation, physical Markov spacing, and step size under separate message and environment-clock budgets. An exact Gaussian subclass exposes the opposing variance, mixing, and horizon effects and gives a correlation-limited speedup ceiling.
- **Lyapunov-Certified Joint Resource Control:** We derive a finite-time Lyapunov certificate for affine delayed

temporal-difference recursions with conditional Markov bias and dependent random Jacobian–innovation pairs. The certificate supplies both the convergence theorem and an implementable joint action score, with scalar step search and no matrix inversion or preconditioning. Its moving-policy extension quantifies how critic contraction competes with policy-induced target motion and transfers the tracking bound to advantage error. We further prove that predictable conservative moment and mixing certificates preserve safety and quantify their finite-horizon selection loss.

- **Information-to-Learning Phase Characterization:** We characterize the price of learning dependence. A stopped adaptive change-of-measure identity and a controlled-belief occupation program give a necessary information cost, while a learning-aware gate gives a sufficient finite-budget performance and safety guarantee. On a compact separated class, the two thresholds match up to class constants and explicit oracle and safety scales.
- **Fully Charged Mechanism and Policy-Learning Evidence:** We validate the complete causal chain with fully charged experiments. Controlled studies recover the predicted participation phase, verify the finite-time confidence bounds, demonstrate 54.4 and 10.9 percent learning-value improvements, and reduce held-out parameter-error and discounted-return-estimation AUC by 9.9 and 11.3 percent over a development-selected strong fixed-participation baseline. Held-out MAPPO studies on two cooperative environments select the dependence-matched action in all 32 controller runs and improve equal-mixture team-return AUC by 2.32 and 8.34 percent over the strongest single fixed action. Complete fixed-participation comparisons over $q \in \{1, 2, 4, 8\}$ place both task-specific controllers within a prespecified two-percent noninferiority margin of their regime-wise fixed envelopes.

The remainder of the paper develops this argument as follows. Section II defines the Markov policy-evaluation model, budgets, and risks. Section III develops the correlation and resource geometry. Section IV derives the Lyapunov controller. Section V establishes the learning-aware adaptation theory. Section VI describes implementation and the stochastic-delay interpretation. Section VII presents the simulations, and Section VIII concludes the paper.

II. MODEL AND FINITE-BUDGET OBJECTIVE

A. Fixed-policy evaluation with multiple actors

Let $(\mathcal{S}, \mathcal{U}, P, r, \gamma)$ be a discounted Markov decision process and let $\pi_{\text{tar}}(u \mid s)$ be a fixed target policy. Under π_{tar} , the state–action process is a Markov reward process with transition kernel P_π and stationary distribution π . For a feature vector $\phi(s) \in \mathbb{R}^d$, the linear value approximation is $V_w(s) = \phi(s)^\top w$. Its projected Bellman fixed point w^* solves

$$Aw^* = c, \quad A = \mathbb{E}_\pi[\phi_t(\phi_t - \gamma\phi_{t+1})^\top], \quad c = \mathbb{E}_\pi[r_t\phi_t]. \quad (2)$$

The target of learning is common across actors; their Markov paths may nevertheless be cross-correlated.

Let S_k denote the physical observation clock at retained learning decision k . The server chooses $q_k \in \{1, \dots, Q\}$ actors and spacing $b_k \in \mathbb{N}$, so $S_k - S_{k-1} = b_k$. Actor i returns $X_{i,k} = (s_{i,S_k}, r_{i,S_k}, s_{i,S_{k+1}})$ and the corresponding temporal-difference tuple. The participating joint chain has invariant distribution π_q and the total-variation certificate

$$\sup_z \|P_q^b(z, \cdot) - \pi_q\|_{\text{TV}} \leq C_q r_q^b, \quad 0 < r_q < 1. \quad (3)$$

The marginal law of every actor is the target-policy law, whereas cross-agent dependence appears in the stationary moments of their aggregate update. This is not a partially observed decision problem: the controller does not infer an execution-time environment state or change π_{tar} . Its uncertainty concerns the dependence and finite-time risk of the training data.

For the identification layer, a Gaussian common-factor subclass makes the dependence parameter explicit. Under instance $j \in \{0, 1\}$,

$$\begin{aligned} C_{S_k} &= \lambda^{b_k} C_{S_{k-1}} + \sqrt{1 - \lambda^{2b_k}} \xi_{j,k}, \\ X_{i,k} &= C_{S_k} + \epsilon_{i,k}, \end{aligned} \quad (4)$$

where $\xi_{j,k} \sim \mathcal{N}(0, \theta_j)$, $\epsilon_{i,k} \sim \mathcal{N}(0, 1)$, and all private innovations are independent. The common process is initialized with $C_{S_0} \sim \mathcal{N}(0, \theta_j)$, so the probe law is stationary. The corresponding same-time correlation is $\rho_j = \theta_j / (1 + \theta_j)$. The model preserves every individual marginal while exposing effective-sample-size saturation caused by common randomness. It is used to prove the information boundary; the convergence theorem below applies to bounded affine temporal-difference observations satisfying Assumption 1.

B. Affine delayed temporal-difference recursion

Let $w^* \in \mathbb{R}^d$ be the projected temporal-difference fixed point and $e_k = w_k - w^*$. For action $a = (q, b, \eta)$ and delays $0 \leq \tau_i \leq D$, the retained recursion is

$$e_{k+1} = e_k - \frac{\eta}{q} \sum_{i=1}^q (H_{i,k} e_{k-\tau_i} - \xi_{i,k}). \quad (5)$$

The rank-one matrix $H_{i,k}$ is the sample temporal-difference Jacobian and $\xi_{i,k}$ is the affine innovation at w^* . For linear TD(0), these quantities are

$$H_{i,k} = \phi_{i,k}(\phi_{i,k} - \gamma \phi'_{i,k})^\top, \quad \xi_{i,k} = \phi_{i,k} \delta_i(w^*), \quad (6)$$

where $\delta_i(w) = r_{i,k} + \gamma \phi'_{i,k}^\top w - \phi_{i,k}^\top w$. The physical chain advances by b transitions before each retained update, which makes the mixing error in (3) a controlled quantity.

Assumption 1 (Certified action domain). *The public sets $\mathcal{Q} \subseteq \{1, \dots, Q\}$ and $\mathcal{B} \subset \mathbb{N}$ are finite, while the step interval $[\eta_{\min}, \eta_{\max}]$ is compact. For each $a = (q, b, \eta)$ in their product: (a) $\|H_{i,k}\| \leq L$ and $\|\xi_{i,k}\| \leq G$ almost surely; (b) the stationary mean matrix $A = \mathbb{E}_{\pi_q} H_{i,k}$ is common across q and has symmetric part at least μI ; (c) $0 \leq \tau_i \leq D$ and the delay profile is known when the score is evaluated; and (d) the mixing and aggregate-moment upper certificates are predictable.*

We write $\mathcal{A} = \mathcal{Q} \times \mathcal{B} \times [\eta_{\min}, \eta_{\max}]$ for this mixed action domain. The finite sets for q and b encode implementable actor counts and admissible environment clocks rather than a continuous relaxation of a discrete system. The controller optimizes η continuously inside each discrete pair and therefore solves the stated mixed discrete–continuous problem exactly up to a chosen scalar-search tolerance.

C. Dual budgets and terminal risk

One retained update with q actors costs $h + q$ message units and b units of the parallel environment clock. A transition taken simultaneously by several actors advances this clock once; the separate message budget charges every participating actor. A feasible trajectory therefore satisfies

$$\sum_k (h + q_k) \leq B_{\text{msg}}, \quad \sum_k b_k + D \mathbf{1}\{K > 0\} \leq B_{\text{env}}. \quad (7)$$

For a stationary action a , let $N_a(B, D)$ be the number of updates usable after charging both budgets and the terminal delay. Explicitly,

$$N_a(B, D) = \left[\min \left\{ \left\lfloor \frac{B_{\text{msg}}}{h + q} \right\rfloor, \left\lfloor \frac{B_{\text{env}} - D}{b} \right\rfloor \right\} \right]_+. \quad (8)$$

The controller minimizes a finite-time certificate $\bar{R}_j(a; B, D)$ satisfying $\mathbb{E}_j \|e_{N_a}\|^2 \leq \bar{R}_j(a; B, D)$. The primary risk is parameter mean-square error. It also controls fixed-policy return estimation because, for a specified initial distribution ν ,

$$\mathbb{E} |J_\nu(w) - J_\nu(w^*)|^2 \leq \|\mathbb{E}_{s \sim \nu} \phi(s)\|^2 \mathbb{E} \|e\|^2, \quad (9)$$

where $J_\nu(w) = \mathbb{E}_{s \sim \nu} [V_w(s)]$.

The Gaussian running-mean learner used in the mechanism study starts from a unit pseudo-observation. For $r = \lambda^b$, its exact risk is

$$R_j^{\text{mean}}(a, N) = \frac{1 + N/q + \theta_j \left[N + 2 \sum_{s=1}^{N-1} (N - s) r^s \right]}{(N + 1)^2}. \quad (10)$$

Equation (10) makes the two opposing effects transparent: increasing q reduces the private-noise term N/q but can reduce N , while increasing b reduces r but consumes environment transitions.

III. CORRELATION AND RESOURCE GEOMETRY

A. A sharp spatial speedup ceiling

The statistical role of participation is exact in a Gaussian location subclass. Let $Y_{i,t} = \vartheta + C_t + \epsilon_{i,t}$ with common variance σ_c^2 , private variance σ_e^2 , and independent rounds. For a fixed participation q , the minimax squared-error risk after T rounds is

$$R_q(T) = \frac{1}{T} \left(\sigma_c^2 + \frac{\sigma_e^2}{q} \right). \quad (11)$$

Writing $\rho = \sigma_c^2 / (\sigma_c^2 + \sigma_e^2)$ gives the speedup relative to one actor,

$$S(q, \rho) = \frac{R_1(T)}{R_q(T)} = \frac{q}{1 + (q - 1)\rho} \leq \min\{q, \rho^{-1}\}. \quad (12)$$

Thus spatial scaling is linear only near independence. For every fixed $\rho > 0$, its benefit saturates while the message cost continues to grow linearly in q .

Proposition 1 (Participation phase under a message budget). *Suppose the environment budget does not bind, the update overhead is h , and a total message budget B_{msg} is used with a fixed q . Ignoring integer rounding, the minimax risk is proportional to*

$$F(q; \rho, h) = (h + q) \left(\rho + \frac{1 - \rho}{q} \right). \quad (13)$$

Its continuous minimizer is

$$q_{\text{cont}}^* = \sqrt{\frac{h(1 - \rho)}{\rho}}, \quad (14)$$

clipped to $[1, Q]$, and an integer optimum is one of its two neighboring feasible participation levels.

Proposition 1 gives a nonheuristic phase boundary. More fixed overhead favors larger batches, whereas stronger common randomness favors smaller participation. The full controller extends this balance to a binding environment clock, Markov spacing, step size, and delay.

B. Temporal spacing and the usable horizon

For the common-factor process in (4), retaining every b th observation changes persistence from λ to $r = \lambda^b$. The integrated correlation term in (10) admits the closed form

$$\sum_{s=1}^{N-1} (N - s) r^s = \frac{r\{(N - 1) - Nr + r^N\}}{(1 - r)^2}. \quad (15)$$

Increasing b decreases this term exponentially, but (8) decreases approximately as $1/b$ when the environment clock binds. Consequently, aggressive thinning can make retained observations nearly independent and still increase terminal error.

For a fixed action, the message-limited and environment-limited horizons are

$$N_a^{\text{msg}} = \left\lfloor \frac{B_{\text{msg}}}{h + q} \right\rfloor, \quad N_a^{\text{env}} = \left\lfloor \frac{B_{\text{env}} - D}{b} \right\rfloor. \quad (16)$$

The active resource is the smaller horizon. This minimum is why optimizing an effective sample size per update, a variance per message, or a mixing time alone can select the wrong action.

C. From local drift to finite-budget value

Let a candidate action have one-block contraction $c(a) < 1$, residual floor $R_*(a)$, and usable horizon N_a . Its certified finite-budget value has the generic form

$$R_*(a) + c(a)^{N_a/m} (R_0 - R_*(a))_+. \quad (17)$$

The first term favors low-variance, well-mixed updates, while the second favors rapid contraction and many affordable updates. The next section derives $c(a)$ and $R_*(a)$ for delayed affine temporal-difference learning rather than postulating this tradeoff.

IV. LYAPUNOV-CERTIFIED JOINT RESOURCE CONTROL

A. Action-dependent one-block drift

The base Lyapunov function is the squared parameter error $V(e) = \|e\|^2$. Delay prevents its one-step drift from closing in e_k alone because the update also contains $e_{k-\tau_i}$. We therefore analyze the observable-independent sliding envelope

$$R_k = \max_{k-2D \leq s \leq k} \mathbb{E}V(e_s), \quad (18)$$

which is a Lyapunov–Razumikhin construction for the lifted history $(e_k, e_{k-1}, \dots, e_{k-2D})$. The history is used in the proof; the online controller needs only the certified coefficients and does not store $2D + 1$ dense matrices.

Define the aggregate stationary moments

$$K_q = \lambda_{\max}(\mathbb{E}_{\pi_q}[\bar{H}_k^\top \bar{H}_k]), \quad \Omega_q = \mathbb{E}_{\pi_q} \|\bar{\xi}_k\|^2, \quad (19)$$

where $\bar{H}_k = q^{-1} \sum_i H_{i,k}$ and $\bar{\xi}_k = q^{-1} \sum_i \xi_{i,k}$. Let $\delta = C_q r_q^b$ and $\mu_\delta = \mu - 2L\delta$. For the delay root-mean-square $\tau_{\text{rms}} = (q^{-1} \sum_i \tau_i^2)^{1/2}$, define

$$\begin{aligned} a_\delta(\eta) &= 1 - \eta\mu_\delta + \eta^2(2K_q + 4L^2\delta), \\ \beta_\delta(\eta) &= \frac{4\eta G^2 \delta^2}{\mu_\delta} + \eta^2(2\Omega_q + 4G^2\delta), \\ h_\tau(\eta) &= 2\eta^4 L^4 \tau_{\text{rms}}^2, \quad g_\tau(\eta) = 2\eta^4 L^2 G^2 \tau_{\text{rms}}^2. \end{aligned} \quad (20)$$

For positive delay, set $\lambda_* = \sqrt{h_\tau/a_\delta}$ and

$$\begin{aligned} c_{\text{aff}} &= (1 + \lambda_*)a_\delta + (1 + \lambda_*^{-1})h_\tau, \\ d_{\text{aff}} &= (1 + \lambda_*)\beta_\delta + (1 + \lambda_*^{-1})g_\tau. \end{aligned} \quad (21)$$

The zero-delay definitions are the continuous limits $c_{\text{aff}} = a_\delta$ and $d_{\text{aff}} = \beta_\delta$. The term $1 - \eta\mu_\delta$ is the useful mean drift. The $\eta^2 K_q$ and $\eta^2 \Omega_q$ terms price multiplicative and additive sampling noise, the δ terms price finite mixing, and the fourth-order terms price stale-state substitution. Every term changes with at least one controlled coordinate of $a = (q, b, \eta)$.

Theorem 1 (Affine delayed Markov Lyapunov bound). *Suppose Assumption 1 holds, $\mu_\delta > 0$, $a_\delta(\eta) > 0$, and $c_{\text{aff}}(a) < 1$. Let $m = 2D + 1$ and $R_k = \max_{k-2D \leq s \leq k} \mathbb{E}\|e_s\|^2$. Then, for every integer $n \geq 0$,*

$$R_{nm} \leq c_{\text{aff}}(a)^n R_0 + \frac{d_{\text{aff}}(a)}{1 - c_{\text{aff}}(a)}. \quad (22)$$

Theorem 1 turns the Lyapunov function $\|e\|^2$ and its delayed sliding envelope into a design objective. Correlation enters K_q and Ω_q , temporal persistence enters δ , and delay enters both the contraction and forcing terms.

Corollary 1 (Finite-horizon return certificate). *Under Theorem 1, let $n = \lfloor N_a(B, D)/(2D + 1) \rfloor$ and let $\psi_\nu = \mathbb{E}_{s \sim \nu} \phi(s)$. Then*

$$\mathbb{E}|J_\nu(w_{N_a}) - J_\nu(w^*)|^2 \leq \|\psi_\nu\|^2 \left[c_{\text{aff}}^n R_0 + \frac{d_{\text{aff}}}{1 - c_{\text{aff}}} \right]. \quad (23)$$

The corollary connects the optimization variable to a reinforcement-learning quantity. The controller does not maximize sampled return directly; it minimizes a certified value-estimation error for the fixed target policy.

The same certificate also tracks a critic target moved by a policy learner. Let w_r^* be the projected fixed point during epoch r , lasting one replacement block $m = 2D + 1$, and let $Z_r = \max_{r m - 2D \leq s \leq r m} \mathbb{E} \|w_s - w_r^*\|^2$.

Theorem 2 (Moving-policy critic tracking). *Suppose every epoch satisfies Assumption 1, possibly with a different certified action a_r , and its coefficients obey $c_{\text{aff}}(a_r) \leq c < 1$ and $d_{\text{aff}}(a_r) \leq d$. If the policy-induced target moves only between epochs and $\|w_{r+1}^* - w_r^*\| \leq v$, then for every $\chi > 0$ satisfying $\bar{c} = (1 + \chi)c < 1$,*

$$Z_n \leq \bar{c} Z_0 + \frac{(1 + \chi)d + (1 + \chi^{-1})v^2}{1 - \bar{c}}, \quad (24)$$

$$\mathbb{E} |\hat{A}_w - A^*|^2 \leq (1 + \gamma)^2 \Phi^2 Z_n, \quad \|\phi(s)\| \leq \Phi. \quad (25)$$

The bound exposes a moving-target phase: policy improvement increases v , while the resource action determines (c, d) . Recomputing the joint score at actor-epoch boundaries therefore maintains a certified critic lag in centralized-training, decentralized-execution policy learning.

B. Joint action selection

The certificate score for action a is

$$\mathcal{U}_j(a; B, D) = R_*(a) + c_{\text{aff}}(a)^{\lfloor N_a(B, D)/m \rfloor} (R_0 - R_*(a))_+, \quad (26)$$

where $R_*(a) = d_{\text{aff}}(a)/(1 - c_{\text{aff}}(a))$. The certified oracle for a known dependence instance is

$$a_j^*(B, D) \in \arg \min_{a \in \mathcal{A}} \mathcal{U}_j(a; B, D). \quad (27)$$

For every pair (q, b) , define the certified scalar domain

$$\mathcal{E}_{q,b} = \{\eta \in [\eta_{\min}, \eta_{\max}] : \mu_\delta > 0, \quad a_\delta(\eta) > 0, \quad c_{\text{aff}}(q, b, \eta) < 1\}. \quad (28)$$

The controller solves

$$\begin{aligned} \eta_{q,b}^* &\in \arg \min_{\eta \in \mathcal{E}_{q,b}} \mathcal{U}_j(q, b, \eta; B, D), \\ (q^*, b^*) &\in \arg \min_{q,b} \mathcal{U}_j(q, b, \eta_{q,b}^*; B, D). \end{aligned} \quad (29)$$

The first problem is a bounded scalar search. The second is exact enumeration of physically discrete actor counts and clocks, not an optimization heuristic.

Proposition 2 (Existence and computation). *If K_q , Ω_q , and the mixing certificates are finite, then $\mathcal{U}_j$ is continuous on every connected component of $\mathcal{E}_{q,b}$ and attains a minimum on its closure after excluding points with $c_{\text{aff}} \geq 1$. With $N_q = |\mathcal{Q}|$, $N_b = |\mathcal{B}|$, and L_η scalar evaluations per search, the certificate cost is $O(N_q N_b L_\eta)$ arithmetic after the moment table is cached. For rank-one temporal-difference samples, updating that table costs $O(qd)$ arithmetic and $O(d)$ working memory per retained block.*

No Hessian inversion, covariance-matrix inversion, or preconditioner is required. The dependence on q is the unavoidable cost of aggregating the selected actor updates.

C. Why minimizing one-step drift is insufficient

A purely myopic drift rule would minimize $c_{\text{aff}}(a)R_k + d_{\text{aff}}(a)$. Such a rule can prefer a large q or b even when its resource cost leaves too few future contractions. Eq. (26) instead composes the drift over the action-dependent horizon in (8). This changes the role of the Lyapunov function from a stability test to a finite-budget control objective.

The transient and noise-floor phases are also explicit. When $R_0 \gg R_*(a)$, the exponential term dominates and a larger stable step can be valuable. Near the floor, the residual term dominates and lower-correlation participation or greater physical spacing can be preferable. The controller balances these phases within one score rather than switching through a separately tuned rule.

V. LEARNING-AWARE DEPENDENCE ADAPTATION

A. Adaptive information lower bound

When θ_j is unknown, the controller may choose a predictable sequence $A_k = (q_k, b_k, \eta_k)$ and stop at a budget-bounded stopping time τ . Rotating X_k into its normalized agent mean $Y_k = q_k^{-1/2} \mathbf{1}^\top X_k$ and orthogonal contrasts shows that only Y_k contributes to the likelihood ratio. Under hypothesis j , a scalar Kalman filter gives the predictive mean $m_{j,k}^-$ and variance $P_{j,k}^-$, with innovation variance $V_{j,k} = 1 + q_k P_{j,k}^-$.

Let $Z_k = (m_{0,k}^-, P_{0,k}^-, m_{1,k}^-, P_{1,k}^-)$. The conditional directional information is

$$\mathcal{I}_{01}(Z_k, A_k) = \frac{1}{2} \left[\log \frac{V_{1,k}}{V_{0,k}} - 1 + \frac{V_{0,k} + q_k (m_{0,k}^- - m_{1,k}^-)^2}{V_{1,k}} \right]. \quad (30)$$

The dependence of information on the resource action is already visible under a fixed probe. Let $T_n(r)$ be the $n \times n$ Toeplitz matrix with entries $[T_n(r)]_{uv} = r^{|u-v|}$, and let $\zeta_1, \dots, \zeta_n$ be its eigenvalues.

Proposition 3 (Exact fixed-probe information). *For a fixed probe (q, b, n) with $r = \lambda^b$, the directional divergence is*

$$\text{KL}(P_0^{q,b,n} \| P_1^{q,b,n}) = \frac{1}{2} \sum_{\ell=1}^n [x_\ell - 1 - \log x_\ell], \quad (31)$$

where

$$x_\ell = \frac{1 + q\theta_0\zeta_\ell}{1 + q\theta_1\zeta_\ell}. \quad (32)$$

The Bhattacharyya exponent used by the upper rule is

$$I_B(q, b, n) = \frac{1}{2} \sum_{\ell=1}^n \log \frac{1 + q(\theta_0 + \theta_1)\zeta_\ell/2}{\sqrt{(1 + q\theta_0\zeta_\ell)(1 + q\theta_1\zeta_\ell)}}. \quad (33)$$

Equations (31)–(33) retain participation, spacing, persistence, probe length, and both dependence hypotheses. They also show why many highly correlated actors need not provide

proportional identification information: q scales a common-mode eigenvalue rather than creating q independent temporal records.

Theorem 3 (Stopped adaptive information identity). *If the action kernel is predictable and common to both hypotheses and τ is bounded by (7), then*

$$\text{KL}(P_0^\pi | \mathcal{F}_\tau \| P_1^\pi | \mathcal{F}_\tau) = \mathbb{E}_0^\pi \sum_{k=1}^{\tau} \mathcal{I}_{01}(Z_k, A_k). \quad (34)$$

The reverse identity holds after exchanging the hypotheses.

The realized action probabilities cancel in the likelihood ratio, while selection affects the occupation distribution of (Z_k, A_k) . Consequently, any rule whose two directional decision errors are at most $\delta < 1/2$ must satisfy

$$\mathbb{E}_j^\pi \sum_{k \leq \tau} \mathcal{I}_{j,1-j}(Z_k, A_k) \geq \text{kl}(1 - \delta, \delta). \quad (35)$$

Together with (7), this gives an occupation-measure lower bound that prices participation, spacing, stopping, and delay without reducing the adaptive Markov experiment to independent action counts.

For a measurable belief-state set G , define the stopped occupation measure

$$\nu_j(G, a) = \mathbb{E}_j^\pi \sum_{k \leq \tau} \mathbf{1}\{Z_k \in G, A_k = a\}. \quad (36)$$

Let $\Delta_j(z, a)$ denote the certified one-step learning opportunity lost by using action a for identification instead of the known-instance learning action.

Proposition 4 (Adaptive opportunity lower bound). *Every predictable, budget-feasible, δ -correct identification policy has expected opportunity cost at least*

$$\inf_{\nu \in \mathcal{M}_j} \int \Delta_j(z, a) \nu(dz, da), \quad (37)$$

where $\mathcal{M}_j$ contains the controlled Kalman-flow constraints, both resource constraints in (7), and the directional information constraint

$$\int \mathcal{I}_{j,1-j}(z, a) \nu(dz, da) \geq \text{kl}(1 - \delta, \delta). \quad (38)$$

The belief-state flow matters because an early choice of (q, b) changes the predictive variance of later observations. Replacing (37) by independent action counts would discard this selection effect and can understate the required sensing cost.

B. Learning-aware upper rule

A fixed probe design $d = (q_d, b_d, n_d)$ consumes $C_d^{\text{msg}} = n_d(h + q_d)$ and $C_d^{\text{env}} = n_d b_d$. Its length is chosen so the exact Bhattacharyya exponent is at least $\log(1/\delta)$, which bounds each directional error by δ . Let B_d^D be the remaining budget after the probe and terminal delay.

The probe is separated from the learning samples and is fully charged. Separation prevents a data-dependent learning

gradient from being reused as if it were an independent correlation certificate. Full charging means that its $q_d n_d$ actor messages, $b_d n_d$ environment-clock units, and terminal delay are removed before any post-probe certificate is evaluated.

For instance j , define the full-budget all-agent certificate $\bar{R}_j^{\text{all}}(B, D)$, the post-probe certificate optimum $\bar{R}_j^*(B_d^D, D)$, and

$$\begin{aligned} C_j^{\text{probe}}(d) &= \bar{R}_j^{\text{all}}(B_d, 0) - \bar{R}_j^{\text{all}}(B, 0), \\ C_j^D(d) &= \bar{R}_j^{\text{all}}(B_d^D, D) - \bar{R}_j^{\text{all}}(B_d, 0), \\ G_j(d) &= \bar{R}_j^{\text{all}}(B_d^D, D) - \bar{R}_j^*(B_d^D, D), \\ W_j(d) &= \max_{\ell \neq j} \left[\bar{R}_j(a_\ell^*; B_d^D, D) - \bar{R}_j^*(B_d^D, D) \right]_+. \end{aligned} \quad (39)$$

The low-instance safety slack is

$$\epsilon_{\text{safe}}(d) = \left[C_0^{\text{probe}}(d) + C_0^D(d) - G_0(d) + \delta W_0(d) \right]_+. \quad (40)$$

The learning-aware controller probes only when

$$C_1^{\text{probe}}(d) + C_1^D(d) + \delta W_1(d) < G_1(d) \quad \text{and} \quad \epsilon_{\text{safe}}(d) \leq \bar{\epsilon}. \quad (41)$$

Otherwise it executes the all-agent fallback without spending a probe. After an admitted probe, it commits to the certified catalogue minimizer for the identified instance.

The all-agent action is a transparent reference policy rather than an uncharged oracle. Any other publicly specified certified fallback can replace it by substituting its risk in (39); the proof is unchanged.

Theorem 4 (Main finite-budget learning-value guarantee). *Suppose Assumption 1 holds for both instances, the probe is feasible under (7), and its directional errors are bounded by δ . Every design admitted by (41) satisfies*

$$\mathbb{E}_j \|e_{\text{LA}}\|^2 \leq \bar{R}_j^*(B_d^D, D) + \delta W_j(d), \quad (42)$$

has strictly smaller certified high-instance risk than the full-budget all-agent action, and has low-instance risk excess at most $\bar{\epsilon}$. If no design is feasible or admitted, the controller returns the all-agent action and incurs no probing cost.

The theorem aligns the decision time with the risk being certified: the action is chosen before its learning block, and every cost in (42) is charged before evaluating the remaining horizon. It also separates three claims that are often conflated. The likelihood exponent certifies identification, the admission inequality certifies positive high-instance learning value, and ϵ_{safe} certifies the declared low-instance tolerance. None follows from the other two alone.

C. Finite-budget phase theorem

Let budgets grow on a public ray $(B_{\text{msg}}(s), B_{\text{env}}(s)) = (\lfloor s\beta_m \rfloor, \lfloor s\beta_e \rfloor)$. Consider a compact class with $0 < \theta_- \leq \theta_j \leq \theta_+$, $|\theta_1 - \theta_0| \geq \Delta_{\min} > 0$, $\lambda \leq 1 - \gamma$, bounded delay, finite $\mathcal{Q}$ and $\mathcal{B}$, and a compact certified step domain. Assume one fixed probe has uniformly positive directional Bhattacharyya information per retained observation, the catalogue risks obey the uniform expansion $\bar{R}_j(a; s) = K_j(a)/s + O(s^{-2})$, and the high-instance coefficient gap between the

all-agent action and the oracle is at least $g_{\min} > 0$. Finally, assume the catalogue's wrong-commit coefficient is uniformly bounded and the declared error ceiling δ_0 leaves at least half of $g_{\min}$ after charging that term. Let B_N be the smallest scale satisfying the two directional information constraints and B_S the smallest scale at which an admitted probe exists.

Theorem 5 (Necessary–sufficient threshold sandwich). *For every fixed admissible safety target and $0 < \delta \leq \delta_0 < 1/2$, there are class constants $C_K, C'_K < \infty$ such that*

$$B_N \leq B_S \leq C_K B_N + C'_K (B_{\text{oracle}} + B_\epsilon + 1). \quad (43)$$

Below B_N , δ -correct identification is impossible, while above B_S the learning-aware controller achieves the guarantees of Theorem 4.

Theorem 5 connects information and learning rather than treating identification as the final objective. The interval $[B_N, B_S)$ is the identifiable-but-not-yet-valuable phase in which information-only probing spends resources that the remaining learner cannot amortize.

The separated-class conditions are substantive. If λ can approach one without an independent mixing certificate, finite samples cannot uniformly distinguish a persistent common factor from its initial condition. If $|\theta_1 - \theta_0|$ can vanish, the required identification budget diverges. The positive theorem therefore states precisely the regime in which learning-aware adaptation is possible rather than hiding these quantities inside an unspecified mixing constant.

VI. IMPLEMENTATION AND STOCHASTIC-DELAY INTERPRETATION

A. End-to-end controller

Algorithm 1 shows the complete decision path. The public catalogue and confidence level are fixed before observations are drawn. Moment and mixing certificates may come from a system specification or a disjoint calibration stream; the likelihood probe is used only to select between prespecified dependence instances.

The computation separates statistics from control. The spatial moment table stores scalar upper bounds on K_q and Ω_q for each feasible participation. The temporal table stores (C_q, r_q) and evaluates $C_q r_q^b$ for each spacing. Once these tables are available, changing a budget or a delay profile requires only scalar arithmetic and bounded search.

B. Predictable confidence interfaces

The theorem requires upper certificates, not plug-in point estimates treated as truth. For a calibration block ending before action selection, let $\hat{K}_q$, $\hat{\Omega}_q$, and $\hat{r}_q$ be estimators and let u_K, u_Ω, u_r be simultaneous error radii. The controller uses

$$K_q^+ = \hat{K}_q + u_K, \quad \Omega_q^+ = \hat{\Omega}_q + u_\Omega, \quad r_q^+ = \min\{1, \hat{r}_q + u_r\}. \quad (44)$$

On the joint coverage event, substituting these upper quantities preserves every certificate inequality. If $r_q^+ = 1$, or if the resulting μ_δ or step domain is invalid, that action is removed

- 1: **Input:** budgets B , delay profile, action sets $\mathcal{Q}, \mathcal{B}$, step interval, risk allowance $\bar{\epsilon}$
- 2: **for** each dependence instance j and pair (q, b) **do**
- 3: form δ , K_q , Ω_q , and the certified set $\mathcal{E}_{q,b}$
- 4: apply upper bounds as a feasibility shield and solve (29) with the predictable central score
- 5: **end for**
- 6: obtain the instance actions a_j^* and their finite-budget risks
- 7: **for** each feasible prespecified probe d **do**
- 8: charge its message, environment, and delay costs
- 9: evaluate C_j^{probe} , C_j^D , G_j , W_j , and ϵ_{safe}
- 10: **end for**
- 11: **if** some probe satisfies (41) **then**
- 12: choose the admitted probe with smallest certified post-probe risk
- 13: observe the charged probe, run the likelihood test, and commit to a_j^*
- 14: **else**
- 15: execute the certified fallback without probing
- 16: **end if**
- 17: **Output:** value parameter and complete resource ledger

Fig. 1. Learning-aware Lyapunov resource control.

rather than optimistically extrapolated. This predictable interface prevents current-block residuals from being counted repeatedly as independent evidence.

The upper quantities need not also rank the actions. They define a robust-feasibility shield and a valid risk envelope; a predictable central score may rank only the actions that pass this shield. This separation prevents conservative uncertainty radii from needlessly shrinking the executed step while retaining the upper risk guarantee.

Corollary 2 (Robust certificate and selection sensitivity). *Suppose $K_q^+ \geq K_q$, $\Omega_q^+ \geq \Omega_q$, and $r_q^+ \geq r_q$ jointly, and form c_a^+, d_a^+ from the upper values using $\lambda_a^+ = \sqrt{h_a/a_a^+}$. Every action satisfying $\mu_{\delta^+} > 0$ and $c_a^+ < 1$ obeys the true recursion $R_{m+1} \leq c_a^+ R_m + d_a^+$. Moreover, if $c_a, c_a^+ \leq \bar{c} < 1$ and $d_a, d_a^+ \leq \bar{d}$, then*

$$|\mathcal{U}_a^+ - \mathcal{U}_a| \leq \left(nR_0 + \frac{\bar{d}}{(1-\bar{c})^2} \right) \Delta c_a + \frac{\Delta d_a}{1-\bar{c}} =: \epsilon_a. \quad (45)$$

Consequently, minimizing the upper score incurs at most $2 \max_a \epsilon_a$ certified risk beyond the exact-certificate oracle.

Here $\Delta c_a = |c_a^+ - c_a|$ and $\Delta d_a = |d_a^+ - d_a|$ admit the explicit moment- and mixing-slack bounds in Appendix A. Thus coverage controls safety exactly, while certificate tightness controls performance continuously.

The confidence mechanism is deliberately modular. Finite-state chains can use spectral or regeneration certificates, while geometrically ergodic continuous-state chains can use an externally supplied mixing upper bound. The controller's guarantee is conditional on the declared simultaneous coverage event, and its failure probability adds directly to the risk statement by total probability.

C. Lyapunov design through a controlled stochastic delay differential equation

The controlled stochastic delay differential equation (SDDE) associated with one fixed resource action is

$$dE(t) = -A_q E(t - \tau(t))dt + \sqrt{\eta} \Sigma_q^{1/2}(E_t) dW(t). \quad (46)$$

Here $\Sigma_q(E_t)$ retains both additive innovation covariance and the state-dependent quadratic variation of random temporal-difference Jacobians. For $V(E) = \|E\|^2$, Itô's formula gives the infinitesimal drift

$$\mathcal{L}_a V(E_t) = -2E(t)^\top A_q E(t - \tau(t)) + \eta \text{tr} \Sigma_q(E_t). \quad (47)$$

Participation controls the spatial covariance in Σ_q , spacing controls the finite-mixing remainder in both terms, and η controls their relative scale. This calculation motivates optimizing a drift-based resource action rather than attaching a stability proof to a variance heuristic.

Eq. (46) is not used to replace the implementable recursion with a continuous approximation. Bounding the delayed cross term in (47) produces the same contraction–forcing decomposition as (20), while Theorem 1 retains finite steps, integer delays, and exact budget rounding. The SDDE therefore supplies the design geometry; the discrete Lyapunov theorem supplies the finite-time guarantee used by the algorithm.

D. Complexity and deployment

For $N_q N_b$ discrete pairs and L_η scalar evaluations, action construction costs $O(N_q N_b L_\eta)$ after calibration. The likelihood recursion is scalar because the orthogonal spatial contrasts cancel under the two common-factor hypotheses. Online aggregation of rank-one temporal-difference samples costs $O(qd)$ arithmetic and $O(d)$ extra memory. The controller is therefore low complexity relative to the $O(qd)$ update it regulates.

The method concerns centralized training of a common fixed-policy value function. After training, policy execution is decentralized and needs no communication because the target policy is already fixed. This separation makes the resource controller a training-layer component that leaves the deployed policy and its execution interface unchanged.

VII. SIMULATIONS

A. Protocol and evaluation questions

The simulations test six predictions in increasing order of algorithmic scope. The first asks whether cross-agent correlation produces the saturation in (12) and moves the resource-optimal participation level. The second asks whether Theorem 1 is finite, empirically calibrated, and responsive to correlation, mixing, and delay when it jointly selects (q, b, η) . The third asks whether the learning-value gate in (41) improves risk over a controller that probes as soon as identification becomes reliable. The fourth asks whether joint control improves the entire resource-indexed convergence trajectory and the induced discounted-return estimate relative to a strong fixed-participation policy. The fifth asks whether a conservative upper-certificate shield preserves safety without sacrificing the

nominal action and whether the finite-horizon score improves terminal risk over myopic and correlation-only rules. The sixth asks whether the same observable dependence decision transfers to centralized-training–decentralized-execution policy learning and improves team return under equal resources.

All studies use paired common random numbers across policies, charge every message and environment-clock unit, retain every prespecified cell, and use independent same-seed reproductions or disjoint confirmation seeds. The Markov temporal-difference task has seven states on a ring, a bounded sinusoidal reward, four stationary-distribution-orthonormal features, and an exact projected fixed point. Cross-agent dependence is induced by shared and private Markov sources while preserving every agent's marginal transition law. The policy-learning studies use multi-agent proximal policy optimization (MAPPO) with parameter-shared actors and a centralized critic [35]. They cover the discrete PettingZoo multi-agent particle environment `simple_spread_v2` [36] and the continuous-control MaMuJoCo `HalfCheetah-v2/2x3` environment introduced with the FACMAC benchmark [37].

Parameter risk is $\|w - w^*\|^2$ normalized by its value before learning. Return-estimation risk is $|V_w(s_0) - V_{w^*}(s_0)|^2$ under the specified initial state and is normalized in the same way. Area under the convergence curve is computed against charged resource fraction, so a policy cannot appear faster by ignoring the samples or messages used to create its updates. Bootstrap comparisons resample paired seeds and report percentile intervals for the ratio of geometric-mean risks.

B. Correlation-limited participation

EXP-007A evaluates $q \in \{1, 2, 4, 8, 16, 32\}$, six cross-agent correlation levels, three delays, three message-equivalent budgets, thirteen step sizes, and 32 paired seeds. All 134,784 rows are finite and budget-valid. At $q = 32$, median effective participation is 30.996 for independent trajectories and 1.111 at correlation 0.9. At the largest evaluated budget and delay eight, the resource-optimal participation moves from $q = 16$ under independence to $q = 1$ under high sharing. Using the endpoint action selected for the opposite correlation regime increases error by factors of approximately 4.29 and 3.13, respectively. These results isolate the statistical and resource mechanism that motivates controlling participation. They also rule out an update-count interpretation of the gain: every endpoint is compared after applying the same message-equivalent budget.

C. Finite-time certificate and joint actions

EXP-010B evaluates twelve (persistence, ρ , D) scenarios with 32 new paired seeds and 1,152 policy runs. The joint controller searches a stability-screened catalogue of $q \in \{1, 2, 4, 8, 16, 32\}$, charged spacings, and scalar steps. All selected actions satisfy $0 < c_{\text{aff}} < 1$, all 1,152 runs are finite and budget-valid, and the largest algebraic discrepancy between the two equivalent drift expressions is 2.22×10^{-16} .

The finite-time bound beats the no-update risk in all twelve scenarios (12/12). Every one-sided 99 percent bootstrap upper mean lies below its corresponding bound, with a maximum

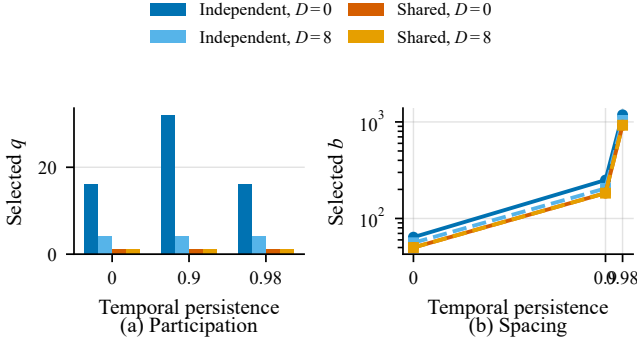


Fig. 2. Joint actions selected by the finite-time Lyapunov certificate in EXP-010B. The left panel reports participation and the right panel reports Markov spacing.

empirical-upper-bound to theorem-bound ratio of 0.305. The median theorem-bound to empirical-mean ratio is 20.77 and the maximum is 228.98, so the certificate is conservative but finite and action-informative.

Fig. 2 shows that the selected participation and spacing respond in the directions predicted by the theory. The controller selects an average participation of 12.67 for independent sources and $q = 1$ in every high-sharing scenario. Delay reduces the selected independent-source participation to $q = 4$, while raising temporal persistence from zero to 0.98 increases the median paid spacing by a factor of 18.49.

D. Learning-value adaptation

EXP-016B uses 96 finite-threshold scenario families, two model layers, eight policies, and 192 independent evaluation seeds. The analysis covers 2,752,512 finite, budget-valid rows across the prespecified scenario grid. The primary estimand is the paired risk difference between probing at B_{id} and applying the learning-value gate in the separation phase.

The learning-aware controller reduces Layer-A risk by 54.43 percent and affine delayed temporal-difference risk by 10.85 percent. The simultaneous one-sided lower bounds on the absolute paired improvements are 0.1612 and 0.0577, respectively. Directional practical improvement occurs in 77 of 96 scenario families. All 96 theorem-facing safety scenarios satisfy $S_{\text{mean}} \leq \epsilon_{\text{safe}}$, and every trajectory obeys both resource budgets.

Fig. 3 separates the conditions under which the value gate is most consequential. The relative risk reductions are 49.12 percent with active delay, 77.69 percent when messages bind, and 43.13 percent when environment transitions bind. The result supports the mechanism in (39): a reliable probe is most costly when it consumes the resource that limits the remaining learner.

E. Convergence and return-estimation curves

TSP-CURVE-001 tests the complete joint controller on the seven-state policy-evaluation problem over a budget of

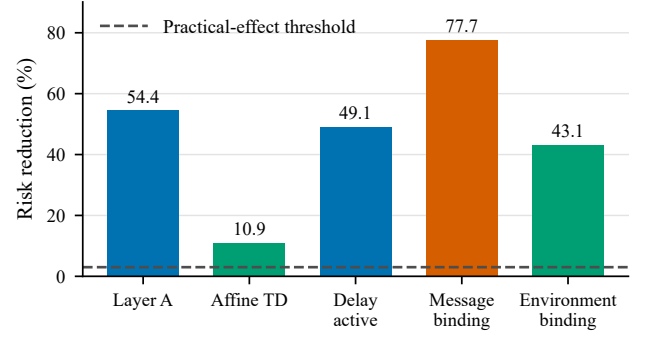


Fig. 3. Relative risk reductions of the learning-aware rule over information-only probing in EXP-016B. The dashed line marks the three-percent practical-effect threshold.

128,000 resource units. The prespecified grid crosses persistence $\{0, 0.9, 0.98\}$, correlation $\{0, 0.9\}$, and delay $\{0, 8\}$, producing twelve cells and 41 charged-resource checkpoints per trajectory. The fixed policies use $q \in \{1, 4, 16, 32\}$; within every cell, each fixed q receives its own certified spacing and step, so the comparison isolates the value of jointly changing participation rather than handicapping its optimizer.

The strong fixed comparator is selected using 16 development seeds and is used without reselection on the confirmation population. It is $q = 4$, which has geometric normalized parameter-error AUC 0.2441 on the development population, compared with 0.3342, 0.4649, and 0.4851 for $q = 1, 16, 32$. The final comparison uses 64 disjoint confirmation seeds and 157,440 finite checkpoint rows.

Relative to fixed $q = 4$, the joint certificate has parameter-error AUC ratio 0.9009 with paired-bootstrap 95 percent interval $[0.8914, 0.9102]$. Its discounted-return-estimation AUC ratio is 0.8868 with interval $[0.8678, 0.9072]$. These correspond to reductions of 9.91 and 11.32 percent. Terminal parameter and return-estimation errors decrease by 26.48 and 21.67 percent, respectively. Both AUC metrics improve in nine of twelve cells and tie in the remaining three; no cell is worse.

Fig. 4 displays the representative persistence-0.9, zero-delay slice. Under independent actor innovations, the joint rule follows the broad-participation action and contracts rapidly. Under correlation 0.9, it follows the low-participation action and avoids the common-noise floor. The figure therefore visualizes the same mechanism predicted by (12) and selected by (29), rather than a post hoc schedule.

F. Transfer across cooperative policy-learning environments

Two held-out MAPPO studies test whether the return-free dependence decision transfers from the affine theory to nonlinear cooperative policy learning. Every method receives the same message and environment budgets, all probe costs reduce the learning horizon, and participation is selected from a disjoint critic-residual probe without evaluation return. Both studies use eight fresh seeds per dependence regime and compare against fixed $q \in \{1, 2, 4, 8\}$. The evaluation separates two claims: noninferiority to the regime-wise fixed

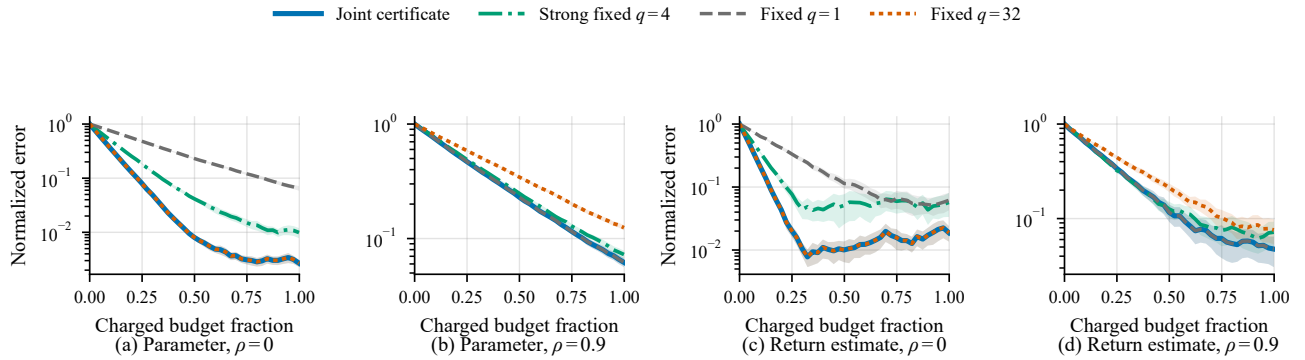


Fig. 4. Held-out resource-indexed convergence in TSP-CURVE-001 for persistence 0.9 and zero delay (64 confirmation seeds; bands are 95 percent confidence intervals). The first two panels show normalized parameter error; the last two show normalized squared error of the fixed-policy discounted-return estimate. The joint Lyapunov certificate changes participation with the dependence regime and improves both learning curves relative to the development-selected strong fixed $q = 4$ comparator.

envelope and improvement over the strongest single fixed action when the dependence regime is unknown.

In `PettingZoo simple_spread_v2`, the controller commits to $q = 8$ in all eight independent-stream runs and $q = 1$ in all eight shared-stream runs. It lies within 0.357 and 0.889 percent of the corresponding regime-wise fixed envelopes and improves equal-mixture team-return AUC by 2.319 percent over the strongest single fixed action, $q = 8$.

In `MaMuJoCo HalfCheetah-v2/2x3`, the task-specific prespecified catalogue is $\{2, 8\}$, while evaluation still includes all four fixed actions. The controller selects $q = 8$ in all eight independent-stream runs and $q = 2$ in all eight shared-stream runs. Relative to the corresponding regime-wise fixed envelopes, its AUC changes are -0.542 and -0.997 percent, both within the prespecified two-percent noninferiority margin. When the regime is unknown, the strongest single fixed action is $q = 8$; against it, the controller improves equal-mixture AUC by 8.341 percent with a one-sided 95 percent lower bound of 1.107 percent, improves seven of eight paired seeds ($p = 0.0352$), and raises final return by 4.637 percent.

Fig. 5 shows the unsmoothed deterministic team-return curves against the maximum charged dual-budget fraction. The probe uses at most 0.768 percent of the message budget, selection overhead is below 10^{-6} of training time, and the two environments show that one Lyapunov learning-value principle supports different task-specific catalogues.

G. Ablations, complexity, and reproducibility

TSP-V3-CERT-SHIELD-002 evaluates the confidence interface with 64 disjoint confirmation seeds and 157,440 check-point rows. The moderate stress inflates K_q and Ω_q by 10 percent and weakens the certified mixing log-rate by 0.05; the strong stress uses 25 percent and 0.10. Both shields retain the nominal action in all twelve cells, all 24 stress-by-cell one-sided 99 percent upper means remain below the robust certificate, and neither shield changes empirical terminal risk.

The same study also isolates the finite-horizon objective. Against one-step myopic drift, terminal parameter risk falls by 12.20 percent; against a correlation-only participation rule,

it falls by 64.22 percent. The respective ratios and paired-bootstrap 95 percent intervals are 0.878 [0.836, 0.923] and 0.358 [0.307, 0.412], both strictly below one.

For a finite catalogue, action evaluation is linear in the number of candidate triples. The exact likelihood update is scalar after the spatial reduction, and the affine certificate uses cached scalar moments and a one-dimensional step search. When temporal-difference Jacobians are rank one, online moment probes require $O(qd)$ arithmetic and $O(d)$ memory, matching the order of aggregating q agent updates.

The five controlled Markov studies ran on CPUs, and their aggregate statistics and curves reproduce byte for byte from the versioned data and analysis code. The convergence study uses a fixed action set and metrics, 16 development seeds for comparator selection, and 64 disjoint confirmation seeds for the reported comparison. The robustness study uses eight development and 64 disjoint confirmation seeds; its five scientific tables reproduce byte for byte. The MPE study comprises 80 complete controller and fixed-catalogue runs across both dependence regimes. The `MaMuJoCo` study adds 80 fresh-seed runs with the same complete comparator set. All 160 policy-learning runs are finite and exactly budgeted, and both analyses reproduce their aggregate curves byte for byte.

The ablations isolate all three coordinates: fixing q removes the correlation response in Fig. 4, fixing b removes the 18.49-fold persistence response, and fixing η ignores the delay-dependent domain in (28). Their gaps distinguish the controller from an adaptive batch-size rule.

VIII. CONCLUSION

Multi-agent Markov policy evaluation requires a resource decision before the next update’s value is known. Our finite-budget Lyapunov certificate jointly chooses participation, physical spacing, and step size; a learning-aware gate measures dependence only when the remaining learner can use it. The theory identifies a spatial speedup ceiling, proves delayed Markov convergence and moving-policy critic tracking, lower-bounds adaptive identification cost, and separates the information and learning-value budget thresholds. Fully

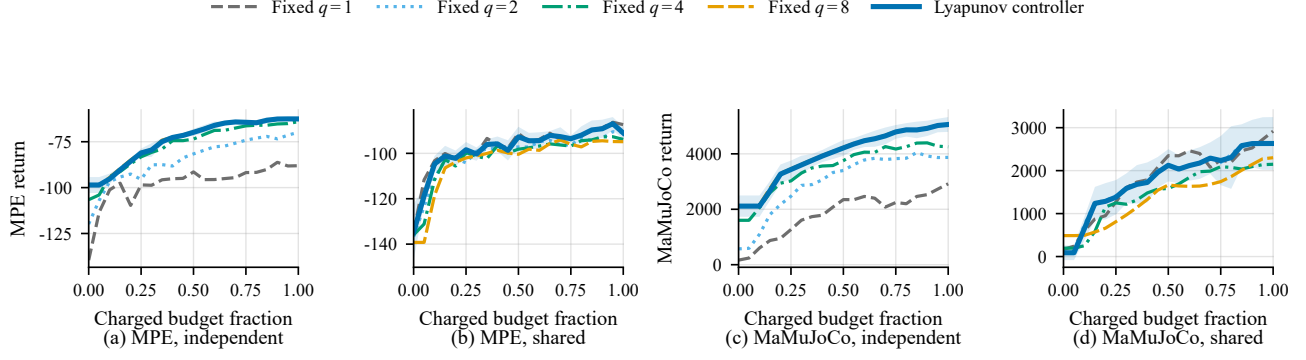


Fig. 5. Fully charged MAPPO team-return learning curves on two cooperative environments with fixed $q \in \{1, 2, 4, 8\}$ and the Lyapunov probe-then-commit controller (eight held-out seeds per regime; shaded bands are controller 95 percent confidence intervals). The controller selects $q = 8$ for independent streams in both environments, $q = 1$ for shared MPE streams, and $q = 2$ for shared MaMuJoCo streams.

charged experiments connect action selection to parameter error, return estimation, and cooperative team-return curves in discrete coordination and continuous-control locomotion. The low-complexity controller uses Lyapunov drift as a design tool and identifies when correlated parallelism is valuable. Routine resource-geometry derivations are provided in the supplementary material.

APPENDIX A

PROOF OF THE AFFINE DELAYED MARKOV BOUND

Fix a stationary action $a = (q, b, \eta)$ and condition on the history immediately before the retained joint observation. All expectations below include both the Markov observation and the random delay profile. Write $\bar{H}_k = q^{-1} \sum_i H_{i,k}$, $\bar{\xi}_k = q^{-1} \sum_i \xi_{i,k}$, and

$$v_k = e_k - \eta(\bar{H}_k e_k - \bar{\xi}_k), \quad r_k = \frac{1}{q} \sum_i H_{i,k} (e_{k-\tau_i} - e_k). \quad (48)$$

Then (5) is $e_{k+1} = v_k - \eta r_k$.

Total-variation duality and stationary centering give

$$\|\mathbb{E}[\bar{\xi}_k | \mathcal{F}_k]\| \leq 2G\delta. \quad (49)$$

Moreover, stationarity and the bounded integrand imply

$$\mathbb{E}[\|\bar{H}_k e - \bar{\xi}_k\|^2 | \mathcal{F}_k] \leq (2K_q + 4L^2\delta)\|e\|^2 + 2\Omega_q + 4G^2\delta. \quad (50)$$

Combining (49) with the strong monotonicity of A and applying Young's inequality with parameter μ_δ yields

$$\mathbb{E}\|v_k\|^2 \leq a_\delta(\eta)R_k + \beta_\delta(\eta). \quad (51)$$

The affine telescoping identity retains the innovation:

$$e_{k-\tau_i} - e_k = \eta \sum_{s=k-\tau_i}^{k-1} \left[\frac{1}{q} \sum_j H_{j,s} e_{s-\tau_j} - \bar{\xi}_s \right]. \quad (52)$$

Jensen's inequality and the almost-sure bounds in Assumption 1 give

$$\mathbb{E}\|r_k\|^2 \leq 2\eta^2 L^4 \tau_{\text{rms}}^2 R_k + 2\eta^2 L^2 G^2 \tau_{\text{rms}}^2. \quad (53)$$

For any $\lambda > 0$,

$$\|v_k - \eta r_k\|^2 \leq (1 + \lambda)\|v_k\|^2 + (1 + \lambda^{-1})\eta^2\|r_k\|^2. \quad (54)$$

Substituting (51) and (53) gives $\mathbb{E}\|e_{k+1}\|^2 \leq c_\lambda R_k + d_\lambda$. The choice $\lambda = \sqrt{h_\tau/a_\delta}$ minimizes the contraction coefficient and produces (21). For zero delay, $r_k = 0$ and the continuous limiting definitions follow directly.

Let $R_\star = d_{\text{aff}}/(1 - c_{\text{aff}})$. If $R_k \geq R_\star$, every new value entering the sliding envelope is at most $c_{\text{aff}}R_k + d_{\text{aff}} \leq R_k$. After $2D+1$ updates, every value in the old envelope has been replaced, so $R_{k+2D+1} \leq c_{\text{aff}}R_k + d_{\text{aff}}$. If $R_k < R_\star$, the same inequality keeps the enlarged envelope below $R_\star$. Iterating the two cases proves Theorem 1.

A. Proof of Corollary 2

Write $\delta^+ = C_q(r_q^+)^b$. On the coverage event, $\delta^+ \geq \delta$ and monotonicity on $\mu - 2L\delta^+ > 0$ gives $a^+ \geq a$ and $\beta^+ \geq \beta$. Evaluating the delayed Young inequality at the shared $\lambda^+ = \sqrt{h/a^+}$ proves the upper recursion; separately optimized forcing terms need not be monotone. For $\Delta\delta = \delta^+ - \delta$, $\Delta K = K_q^+ - K_q$, and $\Delta\Omega = \Omega_q^+ - \Omega_q$,

$$\begin{aligned} \Delta a &= 2\eta^2 \Delta K + (2\eta L + 4\eta^2 L^2) \Delta\delta, \\ \Delta\beta &= 2\eta^2 \Delta\Omega + 4\eta^2 G^2 \Delta\delta \\ &\quad + 4\eta G^2 \left[\frac{(\delta^+)^2}{\mu - 2L\delta^+} - \frac{\delta^2}{\mu - 2L\delta} \right], \\ \Delta c &= (\sqrt{a^+} + \sqrt{h})^2 - (\sqrt{a} + \sqrt{h})^2. \end{aligned} \quad (55)$$

For $h > 0$,

$$\begin{aligned} \Delta d &\leq (1 + \sqrt{h/a^+}) \Delta\beta + \sqrt{h} \left(\frac{1}{\sqrt{a}} - \frac{1}{\sqrt{a^+}} \right) \beta \\ &\quad + \frac{\sqrt{a^+} - \sqrt{a}}{\sqrt{h}} g, \end{aligned} \quad (56)$$

and $\Delta d = \Delta\beta$ when $h = 0$. For $U(c, d) = d/(1-c) + c^n(R_0 - d/(1-c))_+$, $|\partial_d U| \leq (1-\bar{c})^{-1}$ and $|\partial_c U| \leq nR_0 + \bar{d}(1-\bar{c})^{-2}$. The mean-value bound proves (45); applying it to the upper- and exact-score minimizers gives $U_{\bar{a}} \leq U_{a^*} + 2\max_a \varepsilon_a$.

B. Return transfer and arbitrary terminal indices

Let $n = \lfloor N_a / (2D + 1) \rfloor$. The envelope argument keeps every later incomplete-block iterate below $\max\{R_{n(2D+1)}, R_\star\}$, which is the truncated-transient form in (26). Since $J_\nu(w) - J_\nu(w^\star) = \psi_\nu^\top e$ for $\psi_\nu = \mathbb{E}_{s \sim \nu} \phi(s)$, Cauchy–Schwarz and the terminal envelope prove Corollary 1.

C. Proof of Theorem 2

Holding $w_r^\star$ fixed for one replacement block and then applying Young’s inequality to the target shift gives

$$\begin{aligned} \tilde{Z}_{r+1} &\leq c_{\text{aff}}(a_r)Z_r + d_{\text{aff}}(a_r) \leq cZ_r + d, \\ Z_{r+1} &\leq (1 + \chi)cZ_r + (1 + \chi)d + (1 + \chi^{-1})v^2. \end{aligned} \quad (57)$$

Iteration proves (24) whenever $\bar{c} = (1 + \chi)c < 1$. Finally, $\hat{A}_w - A^\star = \{\gamma\phi(s') - \phi(s)\}^\top (w - w_n^\star)$ and $\|\gamma\phi(s') - \phi(s)\| \leq (1 + \gamma)\Phi$; Cauchy–Schwarz proves (25).

D. Existence and complexity of the mixed controller

For fixed (q, b) , every coefficient and (26) is continuous in η on each certified component, including the zero-delay limit; the extreme-value theorem gives existence on its closed feasible portion. The $N_q N_b$ scalar searches cost $O(N_q N_b L_\eta)$ evaluations. For rank-one $H_{i,k}$, feature inner products avoid dense matrices, so calibration costs $O(qd)$ arithmetic and $O(d)$ memory, proving Proposition 2.

APPENDIX B

CORRELATION-LIMITED MINIMAX RISK

For one round of the Gaussian location subclass, the agent average has variance $\sigma_c^2 + \sigma_e^2/q$. The T round averages are independent and Gaussian with known variance, so their sample mean is minimax for squared loss over the unrestricted location family and has risk $T^{-1}(\sigma_c^2 + \sigma_e^2/q)$. Dividing the single-agent risk by this value gives (12). Since $1 + (q-1)\rho \geq q\rho$ for $\rho > 0$, the speedup is at most ρ^{-1} .

APPENDIX C

PROOF OF THE STOPPED ADAPTIVE INFORMATION IDENTITY

Condition on the pre-action history and realized action A_k . An orthogonal rotation maps X_k to $Y_k = q_k^{-1/2} \mathbf{1}^\top X_k$ and hypothesis-invariant standard-normal contrasts. Because the predictable action kernels cancel, the stopped log likelihood is the sum of scalar Kalman innovation ratios, where $Y_k \sim \mathcal{N}(\sqrt{q_k} m_{j,k}^-, V_{j,k})$. Conditional expectation and Tonelli’s theorem give (30) and (34); exchanging hypotheses gives the reverse identity.

For a decision $\hat{J}$, data processing gives

$$\text{KL}(P_j^\pi | \mathcal{F}_\tau \| P_{1-j}^\pi | \mathcal{F}_\tau) \geq \text{kl}(\mathbb{P}_j\{\hat{J} = j\}, \mathbb{P}_{1-j}\{\hat{J} = j\}). \quad (58)$$

If both directional errors are at most δ , monotonicity of binary relative entropy yields (35).

APPENDIX D

OCCUPATION-MEASURE OPPORTUNITY LOWER BOUND

Let $\Delta_j(z, a) \geq 0$ be the one-step opportunity loss relative to the instance- j certified oracle. For a stopped predictable policy, define

$$\nu_j(G, a) = \mathbb{E}_j^\pi \sum_{k \leq \tau} \mathbf{1}\{Z_k \in G, A_k = a\}. \quad (59)$$

Every δ -correct policy induces an occupation measure satisfying the controlled Kalman flow constraints,

$$\begin{aligned} \int \mathcal{I}_{j,1-j}(z, a) \nu_j(dz, da) &\geq \text{kl}(1 - \delta, \delta), \\ \int (h + q(a)) \nu_j(dz, da) &\leq B_{\text{msg}}, \\ \int b(a) \nu_j(dz, da) + D \mathbb{P}_j(\tau > 0) &\leq B_{\text{env}}. \end{aligned} \quad (60)$$

Let T_a be the belief transition and ζ_j the terminal belief law. For every bounded measurable f , predictability and stopped telescoping give

$$\begin{aligned} \int f(z) \zeta_j(dz) - f(z_1) &= \int \left[\int f(z') T_a(z, dz') \right. \\ &\quad \left. - f(z) \right] \nu_j(dz, da). \end{aligned} \quad (61)$$

Moreover,

$$\mathbb{E}_j^\pi \sum_{k \leq \tau} \Delta_j(Z_k, A_k) = \int \Delta_j(z, a) \nu_j(dz, da). \quad (62)$$

Equations (34), (7), and (61) place every admissible occupation measure in the controlled feasible set. Taking the infimum of its exact loss over that larger set proves Proposition 4.

APPENDIX E

PROOF OF THE MAIN LEARNING-VALUE GUARANTEE

Condition on whether the likelihood test identifies instance j correctly. On the correct event, the post-probe risk is at most $\bar{R}_j^\star(B_d^D, D)$. On the wrong event, its excess over the correct catalogue optimum is at most $W_j(d)$. Since the wrong-event probability is at most δ , total expectation proves (42).

Subtracting the full-budget all-agent risk and using (39) gives

$$\mathbb{E}_1 \|e_{\text{LA}}\|^2 - \bar{R}_1^{\text{all}}(B, D) \leq C_1^{\text{probe}} + C_1^D - G_1 + \delta W_1 < 0. \quad (63)$$

The same decomposition on instance zero is bounded above by $\epsilon_{\text{safe}}(d) \leq \bar{\epsilon}$. The fallback statement follows directly from the strict admission rule.

APPENDIX F

PROOF OF THE THRESHOLD SANDWICH

Write $d_\delta = \text{kl}(1 - \delta, \delta)$. Compact separation and $\lambda \leq 1 - \gamma$ upper-bound every feasible observation’s directional information by $i_+ < \infty$; positive action cost and (35) therefore imply $B_N \geq c_N d_\delta$. The fixed probe has information at least $i_- > 0$, so $n_\delta = \lceil i_-^{-1} \log(1/\delta) \rceil$ observations suffice and cost at most $c_p n_\delta + D_{\text{max}}$. Continuity of $\log(1/\delta)/d_\delta$ gives $n_\delta \leq c_0 + c_1 d_\delta \leq c_0 + c_2 B_N$. Uniformly along the budget

ray, every stationary catalogue risk is $K_j(a)/s + O_K(s^{-2})$, so probing perturbs it by at most $L_K(n_\delta + D_{\max} + 1)/s^2$. The high-instance gain is $g_{\min}/s - O_K(s^{-2})$ and the wrong-commit term consumes at most half, hence admission holds for $s \geq c_3 B_N + c_4(B_{\text{oracle}} + 1)$. Low-instance differences are $O_K(s^{-1})$ and satisfy safety once $s \geq B_e$. Taking the maximum proves the upper bound in (43); its lower bound follows because every admitted δ -correct probe obeys both information constraints.

APPENDIX G SECONDARY SDDE INTERPRETATION

The controlled discrete recursion has the local stochastic delay differential equation representation

$$dE(t) = -\frac{1}{q} \sum_{i=1}^q A_i E(t - \tau_i(t)) dt + \sqrt{\eta} \Sigma_q^{1/2}(E_t) dW(t). \quad (64)$$

The diffusion retains the Jacobian-induced state-dependent variation, bounded near the fixed point by $K_q \|E_t\|^2$; this explains the correlation-dependent noise floor and stability margin, while the discrete recursion supplies the finite-time guarantee.

REFERENCES

- [1] R. S. Sutton, "Learning to predict by the methods of temporal differences," *Mach. Learn.*, vol. 3, no. 1, pp. 9–44, Aug. 1988.
- [2] J. N. Tsitsiklis and B. V. Roy, "An analysis of temporal-difference learning with function approximation," *IEEE Trans. Autom. Control*, vol. 42, no. 5, pp. 674–690, May 1997.
- [3] H. Robbins and S. Monro, "A stochastic approximation method," *Ann. Math. Statist.*, vol. 22, no. 3, pp. 400–407, Sep. 1951.
- [4] B. T. Polyak and A. B. Juditsky, "Acceleration of stochastic approximation by averaging," *SIAM J. Control Optim.*, vol. 30, no. 4, pp. 838–855, Jul. 1992.
- [5] J. Bhandari, D. Russo, and R. Singal, "A finite time analysis of temporal difference learning with linear function approximation," in *Proc. 31st Conf. Learn. Theory (COLT)*, 2018, pp. 1691–1692.
- [6] R. Srikant and L. Ying, "Finite-time error bounds for linear stochastic approximation and TD learning," in *Proc. 32nd Conf. Learn. Theory (COLT)*, 2019, pp. 2803–2830.
- [7] W. Mou, A. Pananjady, M. J. Wainwright, and P. L. Bartlett, "Optimal and instance-dependent guarantees for Markovian linear stochastic approximation," in *Proc. 35th Conf. Learn. Theory (COLT)*, 2022, pp. 2060–2061.
- [8] D. Paulin, "Concentration inequalities for Markov chains by Marton couplings and spectral methods," *Electron. J. Probab.*, vol. 20, no. 79, pp. 1–32, Sep. 2015.
- [9] H. Gupta, R. Srikant, and L. Ying, "Finite-time performance bounds and adaptive learning rate selection for two time-scale reinforcement learning," in *Adv. Neural Inf. Process. Syst.*, vol. 32, 2019.
- [10] T. T. Doan, S. T. Maguluri, and J. Romberg, "Finite-time analysis of distributed TD(0) with linear function approximation on multi-agent reinforcement learning," in *Proc. 36th Int. Conf. Mach. Learn. (ICML)*, 2019, pp. 1626–1635.
- [11] K. Zhang, Z. Yang, H. Liu, T. Zhang, and T. Başar, "Fully decentralized multi-agent reinforcement learning with networked agents," in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, 2018, pp. 5872–5881.
- [12] S. Khodadadian, P. Sharma, G. Joshi, and S. T. Maguluri, "Federated reinforcement learning: Linear speedup under Markovian sampling," in *Proc. 39th Int. Conf. Mach. Learn. (ICML)*, 2022, pp. 10997–11057.
- [13] J. Woo, G. Joshi, and Y. Chi, "The blessing of heterogeneity in federated Q-learning: Linear speedup and beyond," in *Proc. 40th Int. Conf. Mach. Learn. (ICML)*, 2023, pp. 37157–37216.
- [14] S. Salgia and Y. Chi, "The sample-communication complexity trade-off in federated Q-learning," in *Adv. Neural Inf. Process. Syst.*, vol. 37, 2024, pp. 39694–39747.
- [15] G. Lan, H. Wang, J. Anderson, C. Brinton, and V. Aggarwal, "Improved communication efficiency in federated natural policy gradient via ADMM-based gradient updates," in *Adv. Neural Inf. Process. Syst.*, vol. 36, 2023, pp. 59873–59885.
- [16] Y. Fraboni, R. Vidal, L. Kamení, and M. Lorenzi, "Clustered sampling: Low-variance and improved representativity for clients selection in federated learning," in *Proc. 38th Int. Conf. Mach. Learn. (ICML)*, 2021, pp. 3407–3416.
- [17] S. De, A. Yadav, D. Jacobs, and T. Goldstein, "Automated inference with adaptive batches," in *Proc. 20th Int. Conf. Artif. Intell. Statist. (AISTATS)*, 2017, pp. 1504–1513.
- [18] M. Cummins, G. D. Er, and M. Muehlebach, "Controlling participation in federated learning with feedback," in *Proc. 7th Annu. Learn. Dyn. Control Conf. (LADC)*, 2025, pp. 174–186.
- [19] J. B. Perazzone, S. Wang, M. Ji, and K. Chan, "Communication-efficient device scheduling for federated learning using lyapunov optimization," *IEEE Trans. Netw.*, vol. 33, no. 4, pp. 1407–1421, Aug. 2025.
- [20] H. Ren, S. Zhao, and S. Ermon, "Adaptive antithetic sampling for variance reduction," in *Proc. 36th Int. Conf. Mach. Learn. (ICML)*, 2019, pp. 5420–5428.
- [21] A. Agarwal and J. C. Duchi, "Distributed delayed stochastic optimization," in *Adv. Neural Inf. Process. Syst.*, vol. 24, 2011.
- [22] X. Lian, Y. Huang, Y. Li, and J. Liu, "Asynchronous parallel stochastic gradient for nonconvex optimization," in *Adv. Neural Inf. Process. Syst.*, vol. 28, 2015.
- [23] Y. Arjevani, O. Shamir, and N. Srebro, "A tight convergence analysis for stochastic gradient descent with delayed updates," in *Proc. 31st Int. Conf. Algorithmic Learn. Theory (ALT)*, 2020, pp. 111–132.
- [24] A. Adibi, N. D. Fabbro, L. Schenato, S. Kulkarni, H. V. Poor, G. J. Pappas, H. Hassani, and A. Mitra, "Stochastic approximation with delayed updates: Finite-time rates under Markovian sampling," in *Proc. 27th Int. Conf. Artif. Intell. Statist. (AISTATS)*, 2024, pp. 2746–2754.
- [25] N. D. Fabbro, A. Adibi, H. V. Poor, S. R. Kulkarni, A. Mitra, and G. J. Pappas, "DASA: Delay-adaptive multi-agent stochastic approximation," in *Proc. IEEE Conf. Decis. Control (CDC)*, Dec. 2024, pp. 3889–3896.
- [26] S. Yu, W. Chen, and H. V. Poor, "Distributed stochastic gradient descent with staleness: A stochastic delay differential equation based framework," *IEEE Trans. Signal Process.*, vol. 73, pp. 1708–1726, Mar. 2025.
- [27] H. Chernoff, "Sequential design of experiments," *Ann. Math. Statist.*, vol. 30, no. 3, pp. 755–770, Sep. 1959.
- [28] S. Nitinawarat and V. V. Veeravalli, "Controlled sensing for sequential multihypothesis testing with controlled Markovian observations and non-uniform control cost," *Sequential Anal.*, vol. 34, no. 1, pp. 1–24, Jan. 2015.
- [29] E. Kaufmann, O. Cappé, and A. Garivier, "On the complexity of best-arm identification in multi-armed bandit models," *J. Mach. Learn. Res.*, vol. 17, no. 1, pp. 1–42, 2016.
- [30] A. Garivier and E. Kaufmann, "Optimal best arm identification with fixed confidence," in *Proc. 29th Conf. Learn. Theory (COLT)*, 2016, pp. 998–1027.
- [31] V. Moulos, "Optimal best Markovian arm identification with fixed confidence," in *Adv. Neural Inf. Process. Syst.*, vol. 32, 2019.
- [32] E. M. Saad, G. Blanchard, and N. Verzelen, "Covariance-adaptive best arm identification," in *Adv. Neural Inf. Process. Syst.*, vol. 36, 2023, pp. 73287–73298.
- [33] F. Vannella, A. Proutiere, and J. Jeong, "Best arm identification in multi-agent multi-armed bandits," in *Proc. 40th Int. Conf. Mach. Learn. (ICML)*, 2023, pp. 34875–34907.
- [34] Y. Wu, R. Shariff, T. Lattimore, and C. Szepesvári, "Conservative bandits," in *Proc. 33rd Int. Conf. Mach. Learn. (ICML)*, 2016, pp. 1254–1262.
- [35] C. Yu, A. Velu, E. Vinitsky, J. Gao, Y. Wang, A. Bayen, and Y. Wu, "The surprising effectiveness of PPO in cooperative multi-agent games," in *Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 24611–24624.
- [36] J. K. Terry, B. Black, N. Grammel, M. Jayakumar, A. Hari, R. Sullivan, L. S. Santos, C. Dieffendahl, C. Horsch, R. Perez-Vicente, N. Williams, Y. Lokesh, and P. Ravi, "PettingZoo: Gym for multi-agent reinforcement learning," in *Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 15032–15043.
- [37] B. Peng, T. Rashid, C. A. S. de Witt, P.-A. Kamienny, P. H. S. Torr, W. Böhm, and S. Whiteson, "FACMAC: Factored multi-agent centralised policy gradients," in *Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 12208–12221.